

Jiménez, Kuvshinov, Peydró, Richter (JF 2026)

Monetary Policy, Inflation, and Crises: Evidence from History and Administrative Data

Contents

1	Big picture and core contributions	3
1.1	Central question	3
1.2	Core empirical claim	3
1.3	Why this is not a purely descriptive point	3
2	Data and key objects	3
2.1	Historical macro panel	3
2.2	Administrative micro evidence (Spain)	3
3	Definitions that unify the empirical narrative	4
3.1	U-shaped monetary rate path	4
3.2	Three-year rate change	4
3.3	Residual vs systematic policy component	4
3.4	Financial red zone (R-zone)	4
4	All models (equations) in the paper	5
4.1	Model 1: Crisis-window event-study regressions (Equation (1))	5
4.2	Model 2: Crisis risk LPM with U-shape interaction (Equation (2))	5
4.3	Model 3: Trilemma instrument for monetary policy changes (Equation (3))	5
4.4	Model 4: Local projections for “crystallization” of vulnerabilities (Equation (4))	6
4.5	Model 5: Spain credit supply and cost of debt (Equation (5))	6
4.6	Model 6: Spain loan-level default risk and U-shape (Equation (6))	7
5	Interpretation of all figures	7
5.1	Figure 1: Select historical crises	7
5.2	Figure 2: Average rates around crises	8
5.3	Figure 3: Crisis-window regression coefficients (Equation (1))	8
5.4	Figure 4: Macrofinancial dynamics around red zones preceded by cuts	8
5.5	Figure 5: Hikes crystallize vulnerabilities (Equation (4))	9

6	Interpretation of all tables (what each table establishes)	9
6.1	Table I: How common is the U-shape <i>before</i> crises and recessions?	9
6.2	Table II: Crisis frequency <i>after</i> different paths	9
6.3	Table III: Crisis risk regression with U-shape interaction (Equation (2))	10
6.4	Table IV: Nonfinancial recessions do not share the same U-shape mechanism	10
6.5	Table V: “Strong” U-shapes driven by residual (non-systematic) policy moves	11
6.6	Table VI: Do long cuts push economies into the financial red zone?	11
6.7	Table VII: Are hikes especially dangerous <i>in</i> the red zone?	11
6.8	Table VIII: The complementarity of U-shapes and red zones	12
6.9	Table IX: Red zones preceded by cuts are the ones that matter	12
6.10	Table X: Banking-sector outcomes under the U-shape	12
6.11	Table XI: Spain micro evidence on credit supply and debt costs (Equation (5))	13
6.12	Table XII: Spain loan-level default risk (Equation (6))	14
7	Synthesis	14
8	Conclusion	15
8.1	Summary: what the paper concludes	15
8.2	Why does the U-shape raise crisis risk?	16
8.3	Takeaway	16

1 Big picture and core contributions

1.1 Central question

The paper asks a policy-relevant, empirically sharp question:

When do monetary policy rate changes raise the risk of systemic banking crises, and how does this differ from “ordinary” business-cycle recessions?

1.2 Core empirical claim

Across a long historical panel, systemic banking crises are disproportionately preceded by a **U-shaped monetary policy rate path**—*extended cuts followed by hikes*—whereas nonfinancial recessions are mainly linked to *rate hikes per se* without a strong dependence on the earlier cut phase.

1.3 Why this is not a purely descriptive point

The U-shape is not only a visual event-study pattern. The paper builds an internally consistent empirical logic:

long cuts $\Rightarrow$ credit/asset-price boom (“red zone”) $\Rightarrow$ hikes in red zone $\Rightarrow$ higher crisis risk.

The paper then strengthens the mechanism with **administrative micro data from Spain**, showing that long rate cuts are associated with **credit supply expansions tilted toward riskier lending**, and that subsequent hikes (especially after earlier cuts) increase **loan-level defaults**.

2 Data and key objects

2.1 Historical macro panel

The historical analysis uses:

- short-term nominal **monetary policy rates** (or closest historical equivalent),
- systemic banking crisis chronologies (baseline: Jordà–Schularick–Taylor, JST; robustness: Baron–Verner–Xiong, BVX),
- macro controls: GDP growth, inflation, etc.,
- additional outcomes: bank profitability, loan losses, bank equity crashes, etc. (Table X sources).

2.2 Administrative micro evidence (Spain)

The paper adds loan-level and firm-level evidence from Spain:

- bank–firm credit commitments (loan-level),

- firm financial accounts to proxy the firm-level cost of debt,
- loan delinquency/default (90+ days past due) as the default measure.

3 Definitions that unify the empirical narrative

3.1 U-shaped monetary rate path

Fix a reference year t . Define cumulative monetary rate changes over two subwindows:

$$\Delta Rate_{t-8,t-3} \equiv Rate_{t-3} - Rate_{t-8}, \quad \Delta Rate_{t-3,t} \equiv Rate_t - Rate_{t-3}.$$

A **U shape** is:

$$\Delta Rate_{t-8,t-3} < 0 \quad \text{and} \quad \Delta Rate_{t-3,t} > 0,$$

i.e., a long cut phase (five years) followed by a shorter hike phase (three years).

3.2 Three-year rate change

A central regressor is:

$$\Delta_{13} Rate_{i,t} \equiv Rate_{i,t} - Rate_{i,t-3},$$

measured in percentage points (ppt).

3.3 Residual vs systematic policy component

The paper distinguishes:

- a **systematic** component of rate changes predicted by macro dynamics,
- a **residual** (discretionary / unexpected) component orthogonal to those dynamics.

Operationally, it residualizes rate changes using contemporaneous and lagged macro controls (GDP growth and inflation, local and global) plus other macro aggregates (investment, consumption, current account), lagged changes in short/long rates, and decade fixed effects. This distinction is crucial for Table V and for the “residual U” concept in Table VIII Panel B.

3.4 Financial red zone (R-zone)

Following Greenwood et al. (2022), the paper defines red zones as episodes where *credit growth and asset price growth are jointly elevated*.

Let y denote one of the components (household credit, business credit, house prices, stock prices). Compute three-year growth $\Delta_{13}y$. Define thresholds:

- **high credit growth:** $\Delta_{13}(\text{credit-to-GDP})$ above the 80th percentile;
- **high asset-price growth:** $\Delta_{13}(\text{real asset prices})$ above the 66.7th percentile.

Then:

- **household R-zone:** household credit + house prices both high;
- **business R-zone:** business credit + stock prices both high;
- baseline indicator: being in *either* household or business R-zone.

4 All models (equations) in the paper

4.1 Model 1: Crisis-window event-study regressions (Equation (1))

The paper starts with an event-window regression (local-projection style):

$$y_{i,t+h} - y_{i,t} = \alpha_{i,h} + \alpha_{d,h} + \beta_h \mathbf{1}\{Crisis_{i,t} = 1\} + \varepsilon_{i,t+h}, \quad h = -7, \dots, 7. \quad (1)$$

Interpretation.

- i indexes countries and t indexes crisis start years.
- $\alpha_{i,h}$ are country fixed effects; $\alpha_{d,h}$ are decade fixed effects.
- β_h traces the average change in y at horizon h relative to the crisis start.

The paper applies (1) to monetary policy rates (raw and residualized), and uses the coefficient path to document the precrisis U-shape.

4.2 Model 2: Crisis risk LPM with U-shape interaction (Equation (2))

To connect the path of policy rates to crisis risk:

$$Crisis_{i,t \rightarrow t+2} = \alpha_i + \beta_1 \Delta_{13}Rate_{i,t} + \beta_2 Cut_{i,t-8 \rightarrow t-3} + \beta_3 \left(\Delta_{13}Rate_{i,t} \times Cut_{i,t-8 \rightarrow t-3} \right) + \gamma' X_{i,t} + u_{i,t}. \quad (2)$$

Interpretation of coefficients.

- β_1 : effect of recent hikes/cuts on crisis risk when there was no long prior cutting phase.
- β_2 : effect of having had long prior cuts (holding recent three-year rate change fixed).
- β_3 : incremental effect of recent hikes conditional on prior long cuts (the U-shape mechanism).

Empirically, the paper emphasizes $\beta_3 > 0$: hikes are especially dangerous after prolonged cuts.

4.3 Model 3: Trilemma instrument for monetary policy changes (Equation (3))

To reduce concerns that rate changes respond to crisis risk, the paper instruments rate changes using the open-economy trilemma logic:

$$TrilemmaIV_{i,t} = \Delta_1 Rate_{b(i),t}^{Residual} \times PEG_{i,t} \times PEG_{i,t-1} \times KOPEN_{i,t}. \quad (3)$$

Objects.

- $b(i)$ is the base country for country i (e.g. anchor currency country).

- $\Delta_1 Rate_{b(i),t}^{Residual}$ is the base-country rate change *residualized* w.r.t. base macro conditions.
- *PEG* indicates a fixed-exchange-rate regime (required in t and $t - 1$).
- *KOPEN* measures capital-account openness (higher $\Rightarrow$ more exposed to base-rate changes).

Identification intuition. Under a peg with open capital markets, domestic short rates must co-move with base short rates for arbitrage reasons, generating quasi-exogenous variation in domestic rates.

4.4 Model 4: Local projections for “crystallization” of vulnerabilities (Equation (4))

To test whether rate hikes trigger stronger reversals when vulnerabilities are already elevated, the paper estimates:

$$\Delta_1 y_{i,t+h} = \alpha_{i,h} + \beta_{1,h} \Delta_1 Rate_{i,t} + \beta_{2,h} \mathbf{1}\{\Delta_{13} y_{i,t} \geq R_z\} + \beta_{3,h} \left(\Delta_1 Rate_{i,t} \times \mathbf{1}\{\Delta_{13} y_{i,t} \geq R_z\} \right) + \gamma' X_{i,t} + \varepsilon_{i,t+h},$$

$$h \in \{1, \dots, 10\}. \quad (4)$$

Interpretation.

- y is one of: household credit, house prices, business credit, stock prices (R-zone components).
- $\mathbf{1}\{\Delta_{13} y_{i,t} \geq R_z\}$ flags a *pre-existing vulnerability* (very high recent growth).
- $\beta_{3,h}$ is the key: if $\beta_{3,h} < 0$ for credit/asset prices, hikes induce larger subsequent declines when vulnerabilities are present.

4.5 Model 5: Spain credit supply and cost of debt (Equation (5))

The administrative data analysis estimates:

$$\Delta_1 y_{b,f,t} = \beta_1 Cut_{t-5 \rightarrow t} + \beta_2 \left(Cut_{t-5 \rightarrow t} \times BankRisk_{b,t-1} \right) + \beta_3 \left(Cut_{t-5 \rightarrow t} \times BankRisk_{b,t-1} \times FirmRisk_{f,t-1} \right) + \gamma'_1 F_{f,t-1} + \gamma'_2 B_{b,t-1} + \gamma'_3 M_t + u_{b,f,t}.$$

$$(5)$$

Interpretation.

- $y_{b,f,t}$ is credit growth (loan-level Δ_1 log commitments) or firm-level cost of debt.
- $Cut_{t-5 \rightarrow t} = 1$ if overnight rates declined over the previous five years (relative to an average benchmark).
- *BankRisk* is proxied by bank NPL ratios; *FirmRisk* by being in real estate/construction.
- The interaction β_3 tests whether credit supply expansions under long cuts are tilted toward risky *bank-firm* pairs.

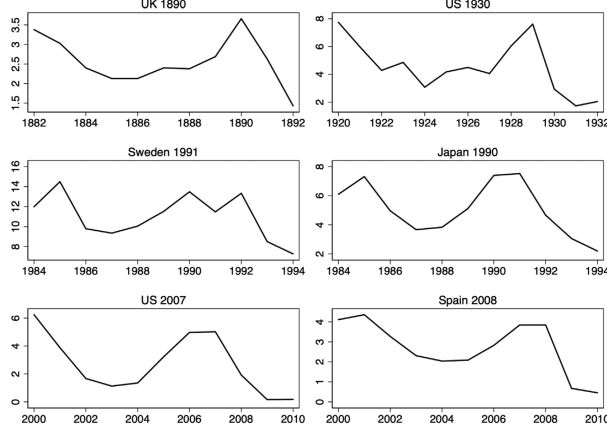


Figure 1. Monetary policy rates around select crisis events. This figure shows the level

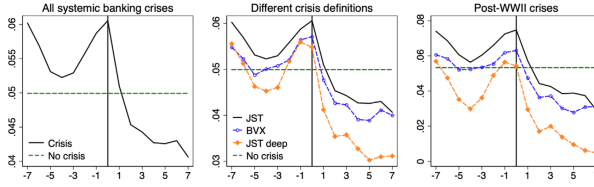


Figure 2. The average level of monetary policy rates around past crises. This figure plots

4.6 Model 6: Spain loan-level default risk and U-shape (Equation (6))

Finally, the paper links subsequent defaults to the U-shape:

$$\begin{aligned} LoanDefault_{f,b,t \rightarrow t+3} = & \beta_1 \Delta_{13}Rate_{t \rightarrow t+3} + \beta_2 Cut_{t-5 \rightarrow t} + \beta_3 \left(\Delta_{13}Rate_{t \rightarrow t+3} \times Cut_{t-5 \rightarrow t} \right) \\ & + \gamma'_1 F_{f,t-1} + \gamma'_2 B_{b,t-1} + \gamma'_3 M_t + u_{f,b,t}. \end{aligned} \quad (6)$$

Interpretation.

- $\beta_1 > 0$: higher rates after origination increase default risk (debt-service channel).
- $\beta_2 > 0$: loans originated after prolonged cuts are riskier (risk-taking / mispricing channel).
- $\beta_3 > 0$: the effect of subsequent hikes is stronger when origination followed prolonged cuts (U-shape risk).

5 Interpretation of all figures

5.1 Figure 1: Select historical crises

What it shows. Time paths of short-term nominal monetary policy rates around selected crisis events. **Main message.** Even in individual high-profile crises, rates often display a U-shaped pattern: cuts years before, then hikes approaching crisis onset.

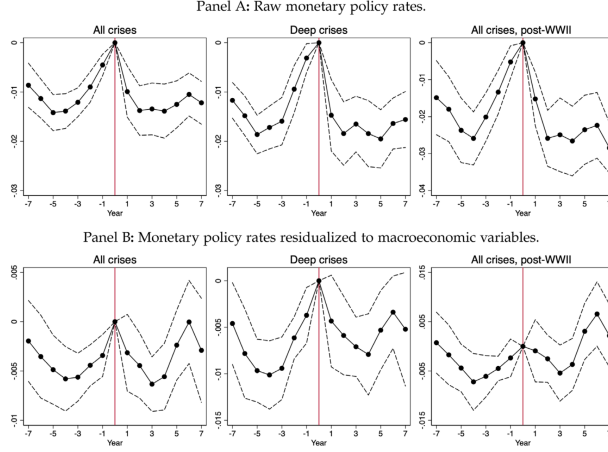


Figure 3. Monetary policy rates—Crisis window regressions. This figure shows regression

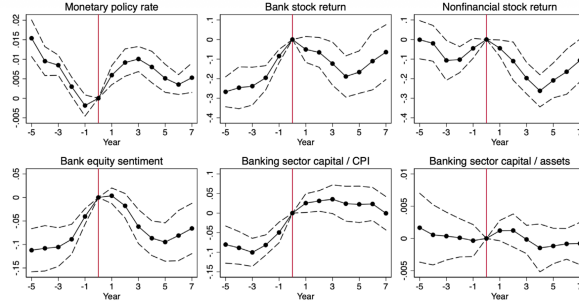


Figure 4. Macrofinancial developments around R-zones preceded by monetary rate cuts

5.2 Figure 2: Average rates around crises

What it shows. Unweighted averages of rate levels around crises for: baseline JST crises, BVX crises, and “deep” crises; also compares with the noncrisis mean. **Main message.** The U-shape is a robust *average* feature and is stronger for deep crises and post-WWII crises.

5.3 Figure 3: Crisis-window regression coefficients (Equation (1))

What it shows. β_h paths from (1) with 90% confidence intervals for raw and residualized rates. **Main message.** The U-shape is statistically precise: rates fall significantly in the early precrisis years and rise significantly in the immediate precrisis years.

5.4 Figure 4: Macrofinancial dynamics around red zones preceded by cuts

What it shows. Event-study coefficients for macro/financial variables around entering an R-zone that was preceded by long rate cuts. **Main message.** The cut phase is followed by a joint boom in credit and asset prices (by construction) and by patterns consistent with credit-supply expansion (e.g., bank equity performance/sentiment dynamics).

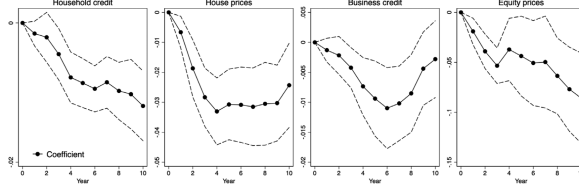


Figure 5. Interaction of monetary rate changes and previous financial red zone vulnerability

Table I
Frequency of Monetary Rate Paths before Crises and Recessions

	(1) All	(2) Deep	(3) Post-WWII	(4) Post-WWII Deep	(5) Unconditional
Panel A: Banking Crises					
U shape (cut, raise)	0.55***	0.68***	0.71***	1.00***	0.27
Raise, raise	0.19	0.14	0.12	0.00	0.24
Raise, cut	0.16	0.11	0.08	0.00	0.26
Cut, cut	0.10	0.08	0.08	0.00	0.23
Panel B: Nonfinancial Recessions					
U shape (cut, raise)	0.34***	0.29	0.33*	0.27	0.27
Raise, raise	0.21	0.21	0.29	0.40	0.24
Raise, cut	0.24	0.24	0.24	0.20	0.26
Cut, cut	0.20	0.26	0.14	0.13	0.23

5.5 Figure 5: Hikes crystallize vulnerabilities (Equation (4))

What it shows. $\beta_{3,h}$ coefficients from (4), capturing the interaction of rate changes with pre-existing vulnerabilities. **Main message.** When credit/asset-price growth has been very high (vulnerable state), rate hikes predict substantially larger subsequent reversals in those variables, consistent with a “crystallization” mechanism.

6 Interpretation of all tables (what each table establishes)

6.1 Table I: How common is the U-shape *before* crises and recessions?

Design. Classify the 8-year rate path before $t+1$ into four bins: U-shape (cut then raise), raise-raise, raise-cut, cut-cut. Report frequencies conditional on crisis and conditional on nonfinancial recessions, plus unconditional.

Key takeaway.

- Unconditionally, the four bins occur with roughly similar frequency (U-shape about 27%).
- Conditional on *banking crises*, the U-shape dominates (about 55% overall, 71% post-WWII).
- Conditional on *deep post-WWII crises*, the U-shape is essentially universal (100% in the table).
- For *nonfinancial recessions*, the U-shape is not uniquely dominant; “raise-raise” becomes relatively more important in deep post-WWII recessions.

6.2 Table II: Crisis frequency *after* different paths

Design. Compute the crisis frequency between t and $t+2$ given the prior path classification over $t-8$ to t . **Key takeaway.** Crises are substantially more likely after a U-shape (about 18% vs 10% unconditional), and the gap is even larger post-WWII and for deep crises.

Table II
Frequencies of Crises after Different Monetary Rate Paths

	(1) Crisis	(2) Deep Crisis	(3) Post-WWII Crisis	(4) Post-WWII Deep Crisis
U shape (cut, raise)	0.18***	0.12***	0.16***	0.14***
Raise, raise	0.09	0.04	0.04	0.01
Raise, cut	0.06	0.02	0.02	0.00
Cut, cut	0.06	0.03	0.03	0.01
Unconditional	0.10	0.05	0.06	0.04

Table III
The Path of Monetary Policy Rates and Banking Crisis Risk

	Dependent Variable: Crisis _{t to t+2}							
	Full Sample				Post-WWII			
	OLS		IV		OLS		IV	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta_3 Rate_t$	0.02** (0.01)	0.01 (0.00)	0.03 (0.02)	0.01 (0.01)	0.01 (0.01)	0.01 (0.01)	0.03 (0.03)	0.01 (0.02)
$Cut Rate_{t-8,t-3}$		0.05 (0.03)		0.04 (0.03)		0.04 (0.03)		0.01 (0.03)
$\Delta_3 Rate_t \times Cut Rate_{t-8,t-3}$		0.03** (0.01)		0.07** (0.03)		0.02** (0.01)		0.08*** (0.03)
Country fixed effects	✓	✓	✓	✓	✓	✓	✓	✓
Controls	✓	✓	✓	✓	✓	✓	✓	✓
Kleibergen-Paap Weak ID			46.69	27.48			55.68	25.91
Observations	1,627	1,627	1,627	1,627	951	951	951	951

6.3 Table III: Crisis risk regression with U-shape interaction (Equation (2))

Design. Linear probability model of crisis occurrence between t and $t+2$ on: $\Delta_{13}Rate_t$, a prior-cut dummy, and their interaction; with rich macro controls and lags. Also IV using trilemma shocks.

Key takeaway.

- The interaction term $\Delta_{13}Rate_t \times Cut_{t-8 \rightarrow t-3}$ is large and statistically significant: *hikes predict crises primarily when they occur after a long cutting phase.*
- IV estimates tend to be larger than OLS, consistent with endogenous policy reactions attenuating OLS.

6.4 Table IV: Nonfinancial recessions do not share the same U-shape mechanism

Design. Replace crisis outcome with nonfinancial recession indicators. **Key takeaway.** Rate increases predict nonfinancial recessions, but the interaction with prior cuts is not robust: the U-shape is not the defining feature for recessions.

Table IV
The Path of Monetary Policy Rates and Nonfinancial Recessions

	Normal Recession _{t to t+2}				Deep Recession _{t to t+2}	
	OLS		IV		OLS	IV
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_3 Rate_t$	0.03*** (0.01)	0.02*** (0.01)	0.06** (0.03)	0.06** (0.03)	0.01*** (0.00)	0.04* (0.02)
$Cut Rate_{t-8,t-3}$		-0.05 (0.04)		-0.08* (0.04)	-0.05* (0.02)	-0.06** (0.03)
$\Delta_3 Rate_t \times Cut Rate_{t-8,t-3}$		0.02 (0.01)		-0.00 (0.04)	-0.01 (0.01)	-0.02 (0.02)
Country fixed effects	✓	✓	✓	✓	✓	✓
Controls	✓	✓	✓	✓	✓	✓
Kleibergen-Paap Weak ID			59.12	31.64		29.80
Observations	1,616	1,616	1,616	1,616	1,616	1,616

Table V
Crisis Frequencies and Systematic versus Residual Monetary Rate Changes

	(1) Crisis	(2) Deep Crisis	(3) Post-WWII Crisis	(4) Post-WWII Deep Crisis
Strong U (residual cut & raise)	0.26***	0.19***	0.27***	0.22***
Moderate U (systematic cut or raise)	0.13	0.08*	0.08	0.07
Raise, raise	0.08	0.02	0.05	0.01
Raise, cut	0.03	0.01	0.02	0.00
Cut, cut	0.06	0.03	0.03	0.01
Unconditional	0.10	0.06	0.07	0.04

Table VI
Monetary Rate Cuts and the Financial Red Zone

	Dependent Variable: $R\text{-Zone}_{t+1 \text{ to } t+3}$							
	$\Delta Rate_{t-5,t}$		Cut $Rate_{t-5,t}$		$\Delta Residual Rate_{t-5,t}$		Large Resid. Cut $_{t-5,t}$	
	OLS (1)	IV (2)	OLS (3)	IV (4)	OLS (5)	IV (6)	OLS (7)	IV (8)
See header	-0.02*** (0.01)	-0.05*** (0.02)	0.08** (0.04)	0.35** (0.14)	-0.02*** (0.01)	-0.07** (0.03)	0.12*** (0.03)	0.49*** (0.17)
Country fixed effects	✓	✓	✓	✓	✓	✓	✓	✓
Controls	✓	✓	✓	✓	✓	✓	✓	✓
Kleibergen-Paap	43.13	54.88	61.94	17.05				
Observations	1,335	1,335	1,335	1,335	1,252	1,252	1,252	1,252

6.5 Table V: “Strong” U-shapes driven by residual (non-systematic) policy moves

Design. Compare crisis frequencies for **strong U** (raw *and* residual cut and raise) vs moderate U and other paths. **Key takeaway.** Strong (residual) U-shapes have much higher crisis frequencies (e.g., about 26% vs 10% unconditional), suggesting discretionary policy moves are especially linked to crisis risk.

6.6 Table VI: Do long cuts push economies into the financial red zone?

Design. Predict entering an R-zone in $t + 1$ to $t + 3$ using monetary cuts over $t - 5$ to t (multiple measures; OLS and IV). **Key takeaway.** Monetary cuts (especially residual/large cuts) strongly raise the probability of entering a red zone, consistent with the “cuts $\Rightarrow$ boom” step.

6.7 Table VII: Are hikes especially dangerous *in* the red zone?

Design. Crisis LPM with an R-zone indicator, a hike indicator, and their interaction (and IV versions). **Key takeaway.** The interaction between being in the R-zone and hiking rates is strongly positive: *hikes in vulnerable states predict crisis risk far more than hikes outside red zones.*

Table VII
Raising Monetary Rates in the Financial Red Zone and Crisis Risk

	Dependent Variable: $Crisis_{t \text{ to } t+2}$					
	All Raises			Residual Raises		Systematic Raises
	OLS (1)	OLS (2)	IV (3)	OLS (4)	IV (5)	OLS (6)
$R\text{-Zone}_{t-3 \text{ to } t-1}$	0.13*** (0.04)	0.03 (0.02)	-0.07 (0.07)	0.07** (0.03)	-0.03 (0.06)	0.12*** (0.04)
$I(\Delta_3 Rate_t \geq 0)$		0.05* (0.03)	-0.01 (0.10)	0.02 (0.03)	-0.03 (0.12)	0.05* (0.03)
$R\text{-Zone} \times I(\Delta_3 Rate \geq 0)$		0.18*** (0.05)	0.38*** (0.15)	0.16** (0.07)	0.44*** (0.16)	0.05 (0.06)
Country fixed effects	✓	✓	✓	✓	✓	✓
Controls	✓	✓	✓	✓	✓	✓
Kleibergen-Paap Weak ID			12.04		11.04	
Observations	1,362	1,362	1,362	1,362	1,362	1,362

Table VIII
U-Shaped Monetary Rate Path, Red Zones, and Crisis Frequencies

	(1)	(2)	(3)	(4)
	Crisis	Deep Crisis	Post-WWII Crisis	Post-WWII Deep Crisis
Panel A: All U Shapes				
U-shaped MP & R-zone	0.36*** (18/50)	0.23*** (12/50)	0.36*** (12/34)	0.30*** (10/34)
U-shaped MP & no R-zone	0.10 (11/117)	0.07 (9/117)	0.06 (3/57)	0.05 (3/57)
No U-shaped MP & R-zone	0.11 (11/99)	0.04 (4/99)	0.06 (4/71)	0.01 (1/71)
No U-shaped MP & no R-zone	0.05 (18/364)	0.03 (10/364)	0.02 (4/220)	0.00 (0/220)
Unconditional	0.09 (58/630)	0.05 (34/630)	0.06 (24/382)	0.04 (14/382)
Panel B: Systematic vs. Residual U Shapes				
Residual U-MP & R-zone	0.45*** (14/30)	0.33*** (10/30)	0.48*** (11/22)	0.40*** (9/22)
Systematic U-MP & R-zone	0.22** (3/15)	0.09 (1/15)	0.12 (1/11)	0.09 (1/11)

Table IX
Raising Monetary Rates in the Financial Red Zone, Previous Cuts, and Crisis Risk

	Dependent Variable: Crisis _t to t+2								
	R-Zone			R-Zone, pre-cut		R-Zone, pre-raise			
	OLS		IV	OLS		IV	OLS		IV
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$R\text{-Zone}_{t-3 \text{ to } t-1}$	0.11*** (0.04)	0.03 (0.03)	-0.09 (0.07)	0.16*** (0.05)	0.04 (0.04)	-0.04 (0.11)	0.02 (0.03)	0.01 (0.03)	-0.06 (0.12)
$I(\Delta_3\text{Rate}_t \geq 0)$		0.05 (0.03)	-0.04 (0.10)		0.05** (0.02)	-0.02 (0.11)		0.10** (0.05)	0.10 (0.15)
$R\text{-Zone}_{t-3 \text{ to } t-1} \times I(\Delta_3\text{Rate}_t \geq 0)$		0.18*** (0.06)	0.42*** (0.16)		0.22*** (0.08)	0.39* (0.24)		0.03 (0.07)	0.17 (0.28)
Country fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓
Controls	✓	✓	✓	✓	✓	✓	✓	✓	✓
Kleibergen-Paap Weak ID			12.99			11.09			3.36
Observations	1,470	1,470	1,470	1,470	1,470	1,470	1,470	1,470	1,470

6.8 Table VIII: The complementarity of U-shapes and red zones

Design. Crisis frequencies by bins combining: (i) U-shape vs not; (ii) R-zone vs not. Panel B splits U-shapes into residual vs systematic. **Key takeaway.**

- Crisis probability is extremely high when **U-shape & R-zone** coincide (about 36% in the main panel).
- U-shape without R-zone and R-zone without U-shape have crisis probabilities much closer to the unconditional mean.
- Residual U-shape & R-zone is the most dangerous combination.

6.9 Table IX: Red zones preceded by cuts are the ones that matter

Design. Split red-zone spells into those preceded by rate cuts vs rate raises; re-estimate interaction models. **Key takeaway.** The hike $\times$ red-zone interaction is strongest for *pre-cut* red zones, consistent with the mechanism: cuts fuel the boom, hikes later trigger the bust/crisis transition.

6.10 Table X: Banking-sector outcomes under the U-shape

Design. Replace crisis outcome in (2) with banking outcomes: profitability, loan losses, bank equity returns, equity crash indicator. **Key takeaway.** U-shapes (especially hikes after prolonged cuts) predict:

- higher bank loan losses (realized credit risk),

Table X
The Path of Monetary Policy Rates and Banking Outcomes

	$\Delta RoE_{i,t+2}$		$\Delta Loan\ loss_{i,t+2}$		$\Delta Other\ inc_{i,t+2}$		$\rho_{Bank\ equity}_{i,t+2}$		$Crash\ Bank\ equity_{i,t+2}$	
	OLS (1)	IV (2)	OLS (3)	IV (4)	OLS (5)	IV (6)	OLS (7)	IV (8)	OLS (9)	IV (10)
Δ_Rate_t	-0.13 (0.16)	-0.02 (0.32)	0.65* (0.34)	1.13*** (0.39)	0.42*** (0.13)	1.14 (0.74)	-0.02 (0.01)	0.02 (0.02)	-0.00 (0.00)	-0.00 (0.01)
$Cut\ Rate_{t-3,t-3}$	-0.07 (0.74)	0.42 (0.65)	-1.13 (1.25)	-1.57** (0.74)	-1.44*** (0.55)	-2.17** (0.89)	-0.04 (0.05)	-0.06 (0.05)	0.04 (0.03)	0.03 (0.03)
$\Delta_Rate_t \times Cut\ Rate_{t-3,t-3}$	-0.83*** (0.25)	-3.14*** (1.02)	1.18*** (0.32)	3.17** (1.45)	0.14 (0.15)	-0.94 (0.75)	-0.03* (0.02)	-0.07* (0.04)	0.02** (0.01)	0.07** (0.03)
Country fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Kleibergen-Paap Weak ID		32.18		17.46		13.19		18.27		27.94
Observations	1,350	1,350	770	770	758	758	1,298	1,298	1,627	1,627

Table XI
Monetary Rate Cuts, Lending Volumes, and Cost of Firm Debt in Spain

Panel A. Dependent Variable: $\Delta \log(Credit)_{i,j,t}$, loan level						
	(1)	(2)	(3)	(4)	(5)	(6)
$Cut_{t-5,t}$	2.54* (1.33)	4.80** (2.18)	3.59** (1.55)			
$Cut_{t-5,t} \times Bank\ NPL\ ratio$			2.62** (1.02)	1.32** (0.54)	1.19** (0.57)	
$Cut_{t-5,t} \times Bank\ NPL\ ratio \times Real\ estate\ firm$					2.12*** (0.58)	2.42* (1.44)
Firm FE	Yes	-	-	-	-	-
Macro controls	Yes	Yes	Yes	-	-	-
Bank controls	Yes	Yes	Yes	Yes	Yes	-
Bank FE	Yes	-	-	-	-	-
Time FE	No	No	No	Yes	Yes	-
Firm $\times$ Bank FE	No	Yes	Yes	Yes	Yes	Yes
Bank $\times$ Time FE	No	No	No	No	No	Yes
Firm $\times$ Time FE	No	No	No	No	No	Yes
Lower-level interactions	-	-	-	-	Yes	Yes
Observations	1.9m	1.9m	1.9m	1.9m	1.9m	1.9m
R^2	0.078	0.187	0.187	0.188	0.188	0.518
Panel B. Dependent Variable: $\Delta Cost\ of\ Debt_{i,t}$, Firm Level						
$Cut_{t-5,t}$	-0.20*** (0.07)	-0.18*** (0.06)				
$Cut_{t-5,t} \times Bank\ NPL\ ratio$		-0.13*** (0.04)	-0.21*** (0.05)	-0.29*** (0.05)	-0.32*** (0.05)	-0.30*** (0.05)
$Cut_{t-5,t} \times Bank\ NPL\ ratio \times Real\ estate\ firm$						-0.00 (0.03)
Macro controls	Yes	Yes	-	-	-	-
Main bank FE	Yes	Yes	Yes	-	-	-
Time FE	No	No	Yes	-	-	-
Main bank $\times$ Time FE	No	No	No	Yes	Yes	Yes
Industry $\times$ Location $\times$ Time FE	No	No	No	No	Yes	Yes
Lower-level interactions	-	-	-	-	-	Yes
Observations	1.2m	1.2m	1.2m	1.2m	1.2m	1.2m
R^2	0.163	0.163	0.169	0.171	0.325	0.327

- worse bank equity performance and higher crash probability,
- patterns consistent with crises operating through credit-risk realization rather than purely interest-rate-risk realizations.

6.11 Table XI: Spain micro evidence on credit supply and debt costs (Equation (5))

Design. Regress bank–firm credit growth and firm cost of debt on long-cut episodes and interactions with bank and firm risk. **Key takeaway.**

- Long monetary cuts are associated with stronger credit growth.
- The credit expansion is stronger for high-NPL (riskier) banks and tilted toward real-estate firms.
- Cost of debt falls more for firms borrowing from weaker banks during cut episodes, consistent with looser credit supply / potential mispricing.

Table XII
The Path of Monetary Rates and Loan-Level Defaults in Spain

	Dependent Variable: $Loan\ default_{t+1\ to\ t+3}$					
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_3 Rate_{t,t+3}$	0.001*** (0.000)	0.002*** (0.000)	0.002*** (0.001)	0.000 (0.001)		
$Cut\ Rate_{t-5,t}$	0.010*** (0.002)	0.011*** (0.001)	0.011*** (0.003)	0.016*** (0.004)		
$\Delta_3 Rate_{t,t+3} \times Cut\ Rate_{t-5,t}$		0.003*** (0.000)	0.005*** (0.001)	0.007*** (0.001)		
$\Delta_3 Rate \times Cut \times Bank\ NPL\ ratio$					0.003*** (0.001)	0.003*** (0.001)
$\Delta_3 Rate \times Cut \times Bank\ NPL \times Real\ estate$						0.005*** (0.001)
Industry $\times$ Location FE	No	No	—	—	—	—
Bank controls	No	No	Yes	Yes	Yes	Yes
Bank FE	No	No	—	—	—	—
Macro controls	Yes	Yes	Yes	Yes	—	—
Time FE	No	No	No	No	Yes	Yes
Firm FE	No	No	—	—	—	—
Firm $\times$ Bank FE	No	No	Yes	Yes	Yes	Yes
Firm controls	No	No	No	Yes	Yes	Yes
Lower-level interactions	—	—	—	—	Yes	Yes
Observations	1.1m	1.1m	1.1m	0.7m	0.7m	0.7m
R^2	0.032	0.032	0.551	0.584	0.585	0.585

6.12 Table XII: Spain loan-level default risk (Equation (6))

Design. Loan default probability over the next three years on post-origination rate changes, pre-origination cut dummy, and interaction (plus risk interactions). **Key takeaway.** Defaults rise when:

- rates increase after origination,
- loans are originated after long cuts,
- and especially when hikes follow long cuts (positive interaction).

The interaction is stronger for weak banks and (in the triple interaction) for real-estate lending.

7 Synthesis

- (1) **Stylized fact:** crises are preceded by a U-shaped rate path (Figures 1–3; Tables I–II).
- (2) **Causal structure:** hikes matter mainly *after* long cuts (Table III + IV via Eq. (2); strengthened with trilemma IV).
- (3) **Mechanism 1 (boom):** long cuts raise the probability of entering a joint credit/asset-price boom state (R-zone; Table VI, Figure 4).
- (4) **Mechanism 2 (crystallization):** hikes are especially destabilizing in that boom state (Tables VII–IX; Eq. (4) + Figure 5).
- (5) **Banking-risk realization:** the U-shape shows up in bank losses and bank equity crashes (Table X).
- (6) **Micro foundation:** in Spain, long cuts are associated with risk-tilted credit supply and lower debt costs, and hikes after cuts increase loan defaults (Tables XI–XII; Eq. (5)–(6)).

8 Conclusion

8.1 Summary: what the paper concludes

This paper concludes that **the *path* of monetary policy matters for financial stability**: systemic banking crises are disproportionately associated with a **U-shaped interest-rate path** (*prolonged cuts followed by subsequent hikes*), and this pattern is not a generic feature of ordinary (nonfinancial) recessions.

The empirical conclusion is supported by two complementary blocks of evidence:

- (i) **A long-run historical panel (17 countries, 150+ years)**. Using annual data back to 1870, the paper documents that the pre-crisis interest-rate dynamics exhibit a U-shape *on average* and also at the level of frequencies:
 - Before banking crises, the U-shape occurs **much more frequently** than other rate paths: **55%** of all crises and **71%** of post-WWII crises are preceded by a U-shaped path, versus an unconditional U-shape frequency of **27%**. Deep post-WWII crises are **always** preceded by a U shape in the sample (**100%**).¹
 - Conditional on observing a U-shape, the probability of a crisis in the next three years is **higher**: the crisis frequency after a U-shape is **18%** versus an unconditional crisis frequency of **10%**.²
- (ii) **State- and path-dependent crisis risk regressions**. In regressions of crisis probability on rate changes, the paper finds that **rate hikes predict crises primarily when they follow a prolonged cutting phase**. This is the central “path dependence” result: hikes are not uniformly destabilizing; they are especially risky when they occur after long easing.
- (iii) **The “residual U” is particularly dangerous**. The crisis association is stronger when the U-shape reflects monetary rate movements *beyond the systematic component* predicted by macro conditions (e.g., growth and inflation). In other words, “strong” U-shapes (raw and residual cuts followed by raw and residual hikes) have the highest crisis frequencies. For example, the crisis frequency after a strong residual U is **26%** (vs **10%** unconditional).³
- (iv) **A micro foundation from administrative data (Spain, post-1995)**. The paper shows that prolonged monetary easing is associated with **credit supply expansion tilted toward risk** (consistent with bank risk-taking and some mispricing), and that subsequent hikes are associated with **higher realized credit risk** (higher loan defaults), especially for weaker banks and ex ante risky borrowers. This micro evidence supports the causal mechanism implied by the historical patterns.

A second key conclusion is comparative: **these U-shape patterns are specific to banking crises and do not characterize nonfinancial recessions in the same way**. For example, while nonfinancial recessions show some association with higher rates, the U-shape is not uniquely

¹See Table I (Panel A).

²See Table II.

³See Table V.

prevalent (e.g., deep post-WWII nonfinancial recessions are relatively more likely to be preceded by “raise–raise” paths).⁴

8.2 Why does the U-shape raise crisis risk?

The core idea is that crises are **built in the easing phase** and **triggered (or revealed) in the tightening phase**.

1) The cut phase builds fragility via a boom state. Prolonged rate cuts lower discount rates and debt-service burdens, encouraging leverage, maturity transformation, and risk-taking in credit markets. In the paper’s language, long easing makes it more likely that the economy enters a **financial “red zone”**: a state in which *credit growth and asset price growth are both unusually high*. This state is not merely “good times”; it is a period of vulnerability accumulation.

2) The hike phase crystallizes vulnerabilities. Once the system is in a vulnerable boom state, hikes can trigger nonlinear dynamics: debt-service burdens rise, collateral values and asset prices can fall, and refinancing becomes harder. This combination produces a sharp deterioration in borrower repayment capacity and bank balance sheets. A key conceptual point from the paper is that the resulting banking stress is driven mainly by **realized credit risk** (loan losses and defaults) rather than only by mechanical interest-rate risk (e.g., duration losses).

3) Why the path matters more than “rates are high” alone. A hike can be contractionary in any state, but it becomes *systemically destabilizing* when it arrives after a long period that has encouraged leverage and risky credit expansion. This is why the paper emphasizes the interaction:

$$\text{hikes} \quad \times \quad (\text{previous prolonged cuts} / \text{boom vulnerabilities}).$$

The U-shape captures exactly this sequencing.

4) Why the paper is informative for policy tradeoffs. An interpretation is that monetary policy has *state-dependent* financial-stability consequences. If vulnerabilities have already accumulated, tightening may be necessary for macro objectives (e.g., inflation), but it can also increase crisis risk unless other tools (e.g., macroprudential policy) reduce fragility beforehand. The paper’s evidence therefore pushes against interpreting monetary policy effects as time-invariant “one coefficient” relationships and instead highlights **path dependence + state dependence** as first-order.

8.3 Takeaway

Banking crises are disproportionately preceded by a U-shaped rate path—prolonged cuts followed by hikes—because easing is associated with credit/asset-price boom states that accumulate fragility, and subsequent tightening crystallizes those vulnerabilities into realized credit losses and systemic stress; this pattern is much weaker for nonfinancial recessions.

⁴See Table I (Panel B).